

Natural Language Processing (NLP) with Python

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Python RegEx

A RegEx, or Regular Expression, is a sequence of characters that forms a search pattern. RegEx can be used to check if a string contains the specified search pattern.

RegEx Module

Python has a built-in package called `re`, which can be used to work with Regular Expressions. Import the `re` module:

```
import re
```

RegEx in Python

When you have imported the `re` module, you can start using regular expressions:

Example

Search the string to see if it starts with "The" and ends with "Spain":

```
import re

txt = "The rain in Spain"
x = re.search("^The.*Spain$", txt)
```

RegEx Functions

The **re** module offers a set of functions that allows us to search a string for a match:

Function	Description
findall	Returns a list containing all matches
search	Returns a Match object if there is a match anywhere in the string
split	Returns a list where the string has been split at each match
sub	Replaces one or many matches with a string

Metacharacters

Metacharacters are characters with a special meaning:

Character	Description	Example
[]	A set of characters	"[a-m]"
\	Signals a special sequence (can also be used to escape special characters)	"\d"
.	Any character (except newline character)	"he..o"
^	Starts with	"^hello"
\$	Ends with	"world\$"
*	Zero or more occurrences	"aix*"
+	One or more occurrences	"aix+"
{}	Exactly the specified number of occurrences	"a{2}"
	Either or	"falls stays"
()	Capture and group	

Special Sequences

A special sequence is a **** followed by one of the characters in the list below, and has a special meaning:

Character	Description	Example
\A	Returns a match if the specified characters are at the beginning of the string	"\AThe"

\b	Returns a match where the specified characters are at the beginning or at the end of a word	r"\bain" r"ain\b"
\B	Returns a match where the specified characters are present, but NOT at the beginning (or at the end) of a word	r"\Bain" r"ain\B"
\d	Returns a match where the string contains digits (numbers from 0-9)	"\d"
\D	Returns a match where the string DOES NOT contain digits	"\D"
\s	Returns a match where the string contains a white space character	"\s"
\S	Returns a match where the string DOES NOT contain a white space character	"\S"
\w	Returns a match where the string contains any word characters (characters from a to Z, digits from 0-9, and the underscore _ character)	"\w"
\W	Returns a match where the string DOES NOT contain any word characters	"\W"
\Z	Returns a match if the specified characters are at the end of the string	"Spain\Z"

Sets

A set is a set of characters inside a pair of square brackets `[]` with a special meaning:

Set	Description
[arn]	Returns a match where one of the specified characters (a , r , or n) are present
[a-n]	Returns a match for any lower case character, alphabetically between a and n
[^arn]	Returns a match for any character EXCEPT a , r , and n
[0123]	Returns a match where any of the specified digits (0 , 1 , 2 , or 3) are present
[0-9]	Returns a match for any digit between 0 and 9
[0-5][0-9]	Returns a match for any two-digit numbers from 00 and 59
[a-zA-Z]	Returns a match for any character alphabetically between a and z , lower

	case OR upper case
[+]	In sets, +, *, ., , (), \$, {} has no special meaning, so [+] means: return a match for any + character in the string

The findall() Function

The `findall()` function returns a list containing all matches.

Example

Print a list of all matches:

```
import re

str = "The rain in Spain"
x = re.findall("ai", str)
print(x)
```

The list contains the matches in the order they are found.

If no matches are found, an empty list is returned:

Example

Return an empty list if no match was found:

```
import re

str = "The rain in Spain"
x = re.findall("Portugal", str)
print(x)
```

The search() Function

The `search()` function searches the string for a match, and returns a [Match object](#) if there is a match.

If there is more than one match, only the first occurrence of the match will be returned:

Example

Search for the first white-space character in the string:

```
import re
```

```
str = "The rain in Spain"
x = re.search("\s", str)

print("The first white-space character is located in position:", x.start())
```

If no matches are found, the value `None` is returned:

Example

Make a search that returns no match:

```
import re

str = "The rain in Spain"
x = re.search("Portugal", str)
print(x)
```

The split() Function

The `split()` function returns a list where the string has been split at each match:

Example

Split at each white-space character:

```
import re

str = "The rain in Spain"
x = re.split("\s", str)
print(x)
```

You can control the number of occurrences by specifying the `maxsplit` parameter:

Example

Split the string only at the first occurrence:

```
import re

str = "The rain in Spain"
x = re.split("\s", str, 1)
print(x)
```

The sub() Function

The `sub()` function replaces the matches with the text of your choice:

Example

Replace every white-space character with the number 9:

```
import re

str = "The rain in Spain"
x = re.sub("\s", "9", str)
print(x)
```

You can control the number of replacements by specifying the `count` parameter:

Example

Replace the first 2 occurrences:

```
import re

str = "The rain in Spain"
x = re.sub("\s", "9", str, 2)
print(x)
```

Match Object

A Match Object is an object containing information about the search and the result.

Note: If there is no match, the value `None` will be returned, instead of the Match Object.

Example

Do a search that will return a Match Object:

```
import re

str = "The rain in Spain"
x = re.search("ai", str)
print(x) #this will print an object
```

The Match object has properties and methods used to retrieve information about the search, and the result:

`.span()` returns a tuple containing the start-, and end positions of the match.

`.string` returns the string passed into the function

`.group()` returns the part of the string where there was a match

Example

Print the position (start- and end-position) of the first match occurrence.

The regular expression looks for any words that starts with an upper case "S":

```
import re

str = "The rain in Spain"
x = re.search(r"\bS\w+", str)
print(x.span())
```

Example

Print the string passed into the function:

```
import re

str = "The rain in Spain"
x = re.search(r"\bS\w+", str)
print(x.string)
```

Example

Print the part of the string where there was a match.

The regular expression looks for any words that starts with an upper case "S":

```
import re

str = "The rain in Spain"
x = re.search(r"\bS\w+", str)
print(x.group())
```

Note: If there is no match, the value `None` will be returned, instead of the Match Object.